Automation

DATA201/422 Data Wrangling

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Plan for lectures

- Monday
  - What and why
  - Orchestration files
  - Automated reports
- Tuesday
  - Configuration files
  - Handling errors
  - Continuous integration
  - Coping with barriers to automation

What and why

What is an automated data science pipeline?

- As much as possible done by code, with minimal human intervention
- Obvious targets for automation: models, reading in data
- Less obvious targets: reports, diagnostics, error handling

Example

- example-automated-pipeline.zip

Benefits of automation

- Accuracy
  - Code more reliable than memory or written instructions
- Speed, efficiency
- Replication
- Lowers costs of good practices
  - E.g. trying alternative specifications
  - E.g. incorporating new data
- Less dependency on individual analysts
  - Good for organizations

Orchestration files

What is an orchestration file?

- A file that runs all the other files in the project
- Typically lives in the ‘project root’
  - Not in the Code folder

Example: R

```
1
2 source("src/01-read-people.R")
3 source("src/02-read-measurements.R")
4 source("src/03-add-variable.R")
5 source("src/04-join-data.R")
6 source("src/05-create-plot.R")
7 quarto::quarto_render("report.qmd")
```

Example: Python

```python
import subprocess

subprocess.run(["Rscript", "src/01-read-people.R"], check=True)
subprocess.run(["Rscript", "src/02-read-measurements.R"], check=True)
subprocess.run(["Rscript", "src/03-add-variable.R"], check=True)
subprocess.run(["Rscript", "src/04-join-data.R"], check=True)
subprocess.run(["Rscript", "src/05-create-plot.R"], check=True)
subprocess.run(["quarto", "render", "report.qmd"], check=True)
```

Makefiles

- 1970s technology (like SQL)
- Disadvantages
  - Slightly weird, old-fashioned formats
- Advantages
  - Open source
  - Fast, reliable
  - Clever rules for updating if a file changes
- AIs make them easier to use

Automated reports

Tools

- LaTeX
- Jupyter notebooks
- RMarkdown (“deprecated”, ie in the process of being replaced)
- Quarto

Quarto example

- `example-automated-pipeline.zip/report.qmd`

Should you put the whole pipeline in an automated report?

- Advantages
  - Simple
  - Combines analysis and explanation in same place
- Disadvantages
  - Can get complex and slow unless project is tiny
- My opinion
  - Don’t do it unless the pipeline is tiny
  - Make the automated report as simple as possible (text plus reading in results calculated elsewhere)